Full Crossing-Block Cauchy Modes

An improvement beyond the reviewed ProveIt ETH proposals
and a faster option using more memory

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Abstract

We review the seven ETH refinement packets placed in ProveIt on October 26, 2026, together with the OpenAI preprint on which they depend and the separately filed Claude manuscript. We give a constructive improvement to the contraction engine. Instead of decomposing one column tile at a time, we eliminate a full crossing orientation block. At valuation level i, its integral separated modes have weights

$$a + 1 - i + 4q - 2s_2(q),$$

with the same multiplicity bound as the existing modes of weights $1 + 3q - 2s_2(q)$. A unit triangular factorization proves the exact Smith profile of the full padded block. The new step is the all-level, group-preserving realization of that profile; the reviewed packets already contain the corresponding local odd-unit Smith calculation.

Under the preprint's normalized Boolean tensor-contraction interface, we derive polynomial-space time

$$(s + \mathcal{L})^{O(1)} 2^{(319.685758 + o(1))s/\sqrt{\log_2 s}}.$$

The strongest explicit combination of the earlier mode basis and schedule family approaches coefficient 368.654018, so the improvement is beyond simply combining those packets. A second construction caches group tensors, forms three block tables, and applies division-free Strassen multiplication. It gives time coefficient 225.597058 and space coefficient 196.199446 on the same scale. Convenient eventual bounds are 320 with polynomial space, or 226 time and 197 space in the exponential coefficients.

These are smaller explicit exponential upper bounds. We do not establish a smaller logarithmic order for unrestricted 3-SAT. The full counting consequence uses the unchanged preprint reductions. Exact kernel and mode-summation prototypes are supplied; an end-to-end fast SAT solver and a new Lean formalization are not supplied.

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1 Outcome and the comparison being made

The useful result is a new integral decomposition that fits the existing recursive algorithm, followed by an optional way to sum its modes with more memory. Neither result assumes the unproved compression hypotheses in the earlier research packets.

Let n denote declared Boolean variables, m input clauses, and k a fixed clause width. The strongest bound in the OpenAI manuscript, rather than its older formalized endpoint, is

$$T_{\mathrm{general}}(n,m) = (n+m)^{O_k(1)} 2^{O_k(n \log\log n/\sqrt{\log n})}, \tag{1}$$

$$T_{\mathrm{sparse}}(n,m) = (n+m)^{O_{k,D}(1)} 2^{O_{k,D}(n/\sqrt{\log n})} \quad (m \le Dn). \tag{2}$$

Both use polynomial space. The manuscript's quantitative appendix is the baseline throughout this note [1, 3, 2].

Our numerical coefficients apply to the normalized contraction size s, not directly to n. The reduction from a formula to a tensor network has additional width-dependent constants. In particular, writing $2^{320n/\sqrt{\log n}}$ as the resulting universal 3-SAT bound would be unjustified.

1.1 Two precise guarantees

Use the preprint's following interface, with its size constant normalized to one. A bipartite Boolean tensor network has at most s edges, polynomially many vertices, maximum degree at most $\left\lceil s^{9/10} \right\rceil$, and description length $\mathcal{L}$. Its entries lie in

$$(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell}), \qquad 1 \le P_0, \ell \le s.$$

Each entry is evaluable in time $(s+\mathcal{L})^b 2^{As^{9/10}}$ and space $(s+\mathcal{L})^b$, for fixed constants A, b. All size and arithmetic bounds below are uniform for fixed interface constants. Bounded exceptional sizes can be treated by exhaustive contraction.

Theorem 1.1 (Polynomial-space improvement)**.** *Using the unchanged deformation, interpolation, packing, and reduction identities of the cited preprint, the normalized network can be contracted deterministically in polynomial space and time*

$$(s+\mathcal{L})^{O(1)} 2^{(C_{\mathrm{ps}}+o(1))s/\sqrt{\log_2 s}}, \qquad C_{\mathrm{ps}} = \frac{\pi\sqrt{4200}}{\ln 2\,(1-3\sqrt{12}/128)} = 319.6857570777\ldots.$$

Consequently the coefficient 320 *is valid for all sufficiently large* s*.*

Theorem 1.2 (A faster option using more memory)**.** *Under the same interface and imported identities, there is a deterministic contraction algorithm with bounds*

$$T \le (s+\mathcal{L})^{O(1)} 2^{(C_{\mathrm{fast}}+o(1))s/\sqrt{\log_2 s}},$$
$$S \le (s+\mathcal{L})^{O(1)} 2^{(C_{\mathrm{mem}}+o(1))s/\sqrt{\log_2 s}},$$

where, for $\omega = \log_2 7$ *and* $\kappa = \pi\sqrt{4200}/\ln 2$*,*

$$C_{\mathrm{fast}} = \kappa\sqrt{\frac{43690 - [1-(\omega-2)^2]14450}{65536}} = 225.5970572839\ldots,$$
$$C_{\mathrm{mem}} = \kappa\sqrt{\frac{29240}{65536}} = 196.1994455176\ldots.$$

Thus coefficients 226 *for time and* 197 *for space are valid eventually.*

Theorem 1.1 preserves polynomial space. Theorem 1.2 deliberately spends subexponential space to improve the time coefficient. The latter is a different resource tradeoff, not a polynomial-space speedup.

1.2 How much improves?

For the earlier explicit schedule $r = 256$ and $u_d = \left\lfloor \lambda/(720 \cdot 12^d) \right\rfloor$, the geometric, existing factorial, and new full-block bases have the following coefficients:

$$648.4410666487\ldots, \qquad 374.3776243833\ldots, \qquad 324.2205333244\ldots.$$

The new basis alone multiplies the leading mode-enumeration coefficient by $\sqrt{3}/2$ relative to the earlier factorial basis [5]. The slightly larger admissible taper scale $1/700$ gives Theorem 1.1.

The earlier schedule paper also optimizes its parameter family [6]. Combining its boundary infimum with the old factorial basis gives approximately 368.6540181, rather than 374.3776244. Our polynomial-space coefficient is below both. This boundary value is an infimum of a specified certificate family, not a lower bound for all possible implementations of the earlier ideas. We compare explicit upper-bound certificates; we have not measured a SAT runtime ratio.

2 Repository review and what is already present

The review started at ProveIt commit `aa6933442eb936962de79ba6d321cd0391c450bd` and includes the subsequent placement of the same packets into `Computability/SubexponentialSAT`. The final review snapshot is

`3c64ee065c708ded485b8b14d9ad6c703a30ab11`.

The relevant arrival commits are `a505bd768bce`, `64b3202ee84f`, and `8afdbbea48d6`; the placement commit is `2b925d8800cd`. The placed payloads were compared byte for byte against the seven authenticated archives and the Claude PDF: all 120 placed payload files matched. The detailed provenance records accompany this note.

The OpenAI preprint directory is identical at the two pins used by these packets, `7ef5c0cc5ca4` and `42907c9e999c`. The ProveIt project README explicitly describes the material as external research that has not been checked in ProveIt [4]. We do not turn that placement into a claim of formal verification.

Packet	Established result or stated proposal	Consequence for this work
Factorial Cauchy modes	Exact integral separation with weight $1 + 3q - 2s_2(q)$; constant improvement inside the exponential.	Primary mode-basis comparator.
Contraction improvements	Exact taper count, smaller precision, explicit coefficient 648.4411, and streamed packet extraction.	Reuse its proved refinements; do not rediscover them.
Precision refinement	Exact recovery losses and taper minimax analysis. Faster structured-mode summation is an unproved hypothesis.	No assumed fast oracle is used here.

Packet	Established result or stated proposal	Consequence for this work
SAT density improvement	Density dependence $n \min\{\log(2+m/n), \log\log n\}/\sqrt{\log n}$.	Compatible with the new contraction engine.
ETH density refinement	Same density improvement; local odd-unit Cauchy Smith valuations $4q - 2s_2(q)$ and other constant refinements.	The local slope-four spectrum is already known in the reviewed material.
Scalar parity refinement	Removes the truncated-V coordinate and reduces the incidence-degree cap to $k(n+a)$ for a tests. A second-contraction speedup remains conditional.	Scalar arithmetic can be retained; our tabulation theorem supplies a different, fully costed tradeoff.
Asymptotic analysis	Restricted method barriers; degree-preserving and linear-wire compression targets are not constructed.	The corresponding smaller-order claims remain open.

The separate 49-page Claude manuscript states, for fixed k, the same relevant asymptotic order in n, with bound $2^{O(\min\{k^2 n \log\log n/\sqrt{\log n}, km/\sqrt{\log m}\})}$ [11]. The $\log\log n$ factor is outside the square root. Its different crossing decomposition is not used in our proof.

Novelty relative to the reviewed corpus. Classical Cauchy elimination, Smith normal form, and Strassen multiplication are not new. The additional mathematical step here is a residue-ordered decomposition of an entire crossing orientation, valid at every retained valuation level and over the actual unramified interpolation ring. It attains the slope-four profile without increasing the mode-family bound or breaking the one-source/one-target group property. The second contribution is an explicit mode-graph evaluation with all oracle, time, and space costs paid. This is a comparison with the reviewed sources, not a global priority claim.

3 The crossing block and its arithmetic

Let $\mathcal{O}$ be the preprint's unramified extension of $\mathbb{Z}_{(2)}$, with fraction field $\mathcal{K}$, and normalize $v_2(2) = 1$. An element of $\mathcal{O}$ is a unit exactly when its reduction modulo two is nonzero. The scalar factors below lie in $\mathcal{O}$; tensor entries can lie in $\mathcal{A} = \mathcal{O}[V]/(V^\ell)$. Scalar integrality therefore also suffices when the tensor ring has nilpotents. All logarithms used for sizes are base two; ln denotes the natural logarithm.

At a recursive call, put $L = 2^a$ and

$$\mathcal{C}(T) = \prod_{v=-L}^{L-1} (T - v), \qquad \alpha = 2^{a+1}, \qquad \Gamma(T) = \frac{\mathcal{C}(T)}{\mathcal{C}'(0)}.$$

For row slot x and column slot y in the same packet, the source kernel is

$$W_{xy}(T) = \frac{\Gamma(T)}{T + x - y}. \tag{3}$$

The cancelled expression is a polynomial. At a retained interpolation node $t \neq 0$, set

$$i = v_2(t) < a, \quad R = 2^i, \quad D = L/R, \quad N = L/2, \quad T = t + 2^{a+1}\zeta, \quad \zeta \in \mathcal{O}.$$

The preprint's integrality argument gives

$$v_2(T) = i, \qquad \Gamma(T) \in 2^{a+1}\mathcal{O}. \tag{4}$$

The second assertion follows directly by taking the factor $T - t$ out of $\mathcal{C}(T)$: every other factor has the valuation of its ordinary nonzero grid difference, and those valuations add to $v_2(\mathcal{C}'(0))$. If ζ is a unit, the valuation is exactly $a + 1$; when $\Gamma = 0$, all crossing entries vanish.

Within a packet with unequal symbols in coordinate i, crossing row and column slots have the same bit in that coordinate. Fix that bit $\varepsilon \in \{0, 1\}$. Define

$$\sigma(j) = (j \bmod R) + 2R \lfloor j/R \rfloor, \qquad 0 \le j < N. \tag{5}$$

The slots in this orientation are $\varepsilon R + \sigma(j)$. The offset cancels in $x - y$, so both orientations have the same scalar difference kernel. All columns in one orientation belong to a single source group, and its crossing rows belong to the other group.

Lemma 3.1 (Denominators have known valuation)**.** *For any $j, k < N$, the element $T + \sigma(j) - \sigma(k)$ is nonzero and has valuation at most i. If $j \not\equiv k \pmod R$, its valuation is $v_2(j - k) < i$. If $j \equiv k \pmod R$, its valuation is i.*

Proof. In the first case, insertion of the zero bit at position i preserves the lowest differing bit; adding T, of valuation i, cannot change it. In the second case, the difference of the two slots is divisible by $2R$, while $T/2^i$ is odd modulo two. In either case the normalized denominator is a unit. This reasoning works for any $\zeta \in \mathcal{O}$, including elements outside the prime subring. □

4 An exact integral decomposition of the whole block

Write $s_h = \sigma(h)$. For $0 \le b \le N$ define

$$P_b(x) = \prod_{h<b} \frac{x - s_h}{T + x - s_h}, \qquad Q_b(y) = \prod_{h<b} \frac{s_h - y}{T + s_h - y}.$$

Empty products are one. For $b < N$ set

$$f_b(x) = \frac{TP_b(x)}{T + x - s_b}, \qquad g_b(y) = \frac{TQ_b(y)}{T + s_b - y}. \tag{6}$$

Lemma 4.1 (Finite Cauchy separation)**.** *For $x, y \in \{s_0, \dots, s_{N-1}\}$,*

$$\frac{1}{T + x - y} = \frac{1}{T} \sum_{b=0}^{N-1} f_b(x) g_b(y). \tag{7}$$

Proof. Put $H_b = P_b(x)Q_b(y)/(T + x - y)$. Cross multiplication gives

$$(T + x - s_b)(T + s_b - y) - (x - s_b)(s_b - y) = T(T + x - y).$$

Consequently

$$H_b - H_{b+1} = \frac{TP_b(x)Q_b(y)}{(T + x - s_b)(T + s_b - y)} = \frac{1}{T} f_b(x) g_b(y).$$

The term H_0 is the desired kernel, and $H_N = 0$ because one numerator factor is $x - x$. Telescoping proves the identity without an infinite series or an approximation. □

For $b = qR + r$, $0 \le r < R$, define

$$E_b = q + v_2(q!) = 2q - s_2(q). \tag{8}$$

The second equality is the elementary binary factorial-valuation identity.

Lemma 4.2 (Two factorial gains)**.** *For every $b, j < N$,*

$$f_b(s_j), g_b(s_j) \in 2^{E_b}\mathcal{O}.$$

Both vanish when $j < b$. At $j = b$, their valuation is exactly E_b.

Proof. If $j < b$, the factor with $h = j$ is zero. Suppose $j \ge b$, and write $j = kR+v$. For a predecessor h with residue different from v modulo R, Lemma 3.1 shows that the numerator and denominator in the relevant factor of P_b or Q_b have the same valuation. Such a factor contributes zero net valuation.

The predecessors with residue v are $h = \ell R + v$ for $\ell = 0, \dots, c-1$, where $c = q$ if $v \ge r$ and $c = q+1$ if $v < r$. Their slot differences are $2R(k-\ell)$ and the denominators have valuation i. Their combined contribution is

$$c + v_2(k(k-1)\cdots(k-c+1)) \ge c + v_2(c!) \ge q + v_2(q!).$$

Here $k \ge c$ follows from $j \ge b$, and the first inequality follows from the integrality of the ordinary integer $\binom{k}{c}$. The remaining factors $T/(T + s_j - s_b)$ and $T/(T + s_b - s_j)$ are integral by Lemma 3.1. At $j = b$ they equal one; the same-residue differences are exactly $2R, \dots, 2Rq$, giving equality in the valuation. Only ordinary integer factorials have been used. No assertion that arbitrary unramified extension elements have integral binomial polynomials is needed. □

Define the integral functions

$$F_b(x) = 2^{-E_b} f_b(x), \qquad G_b(y) = 2^{-E_b} g_b(y),$$

and the target coefficient

$$A_b(x) = \frac{\Gamma(T)}{T}\, 2^{2E_b} F_b(x). \tag{9}$$

Theorem 4.3 (Group-preserving slope-four modes)**.** *For every crossing orientation,*

$$W_{xy}(T) = \sum_{b=0}^{N-1} A_b(x) G_b(y), \qquad G_b(y) \in \mathcal{O}, \tag{10}$$

and, writing $b = qR + r$,

$$A_b(x) \in 2^{w_q}\mathcal{O}, \qquad w_q = a + 1 - i + 2q + 2v_2(q!) = a + 1 - i + 4q - 2s_2(q). \tag{11}$$

There are R such modes per q in each orientation, and two orientations per unequal-symbol packet.

Proof. Substitute the integral normalizations into Lemma 4.1 and multiply by $\Gamma(T)$. Equation (4) supplies valuation at least $a + 1 - i$ for $\Gamma(T)/T$. The remaining contribution is $2E_b$. The index interval $0 \le b < N = R(D/2)$ contains exactly R indices for each $0 \le q < D/2$. □

For a mode b, the source linear form is

$$Z_b(X) = \sum_{\substack{y \text{ a real column} \\ y \text{ in this orientation}}} G_b(y) X_y. \tag{12}$$

It reads the entire half-packet, rather than one tile. That half-packet still lies in one source group. Its target coefficients are used only by rows in the other group. Missing real slots are simply omitted from the sum. The auxiliary pivots s_h need not themselves be real ports.

5 Exact Smith profile and what its optimality means

Let $K_{jk} = 1/(T + s_j - s_k)$ for the full padded orientation. Form matrices $\mathsf{L}_{jb} = F_b(s_j)$ and $\mathsf{U}_{bk} = G_b(s_k)$. Lemma 4.2 says that L is lower triangular, U upper triangular, and both have unit diagonal. Therefore

$$TK = \mathsf{L}\ \mathrm{diag}(2^{2E_b})\,\mathsf{U} \tag{13}$$

is an equivalence by invertible integral matrices. Since E_b is nondecreasing, it is a Smith factorization up to unit diagonal factors.

Corollary 5.1 (The full-block invariant factors)**.** *The Smith valuations of TK are*

$$2E_b = 4q - 2s_2(q), \qquad b = qR + r,$$

so each value indexed by q occurs R times. If ζ is a unit, the Smith valuations of the actual crossing block W are exactly $a + 1 - i + 4q - 2s_2(q)$, with the same multiplicities.

For $i = 0$, this specializes to the odd-unit profile already proved in the density refinement packet [8]. The added fact needed for the algorithm is the full constructive factorization at arbitrary i, with integral source functions supported in one group and no extra mode-family factor. The invariant factors cannot be improved by invertible integral changes of basis. This is a local statement about the full padded kernel. It is not a lower bound for SAT, for general tensor contraction, or for sparse real-port restrictions of this block.

6 Compatibility with the recursive contraction

The source packing produces at most $3M/L$ packets at an M-edge call. By Theorem 4.3, the number of modes of a fixed index q is at most

$$\frac{3M}{L}\,2R = \frac{6M}{D}. \tag{14}$$

This is exactly the old multiplicity envelope. In particular, it has not been multiplied by the number of denominator-valuation classes.

Each original row argument is still its local part plus a linear combination of formal modes. The product of its affine Boolean row factors has the property that the coefficient of a mode monomial Z^β is divisible by $2^{\sum w_q \beta_b}$. At working precision p_t, retain only

$$\sum_b w_{q(b)}\beta_b < p_t. \tag{15}$$

All omitted terms vanish. Extract the targeted coefficients in each packet-group tensor and substitute the source forms from (12). Each mode still has one source and one target, so the same exact factorization holds:

$$F(t + \alpha\zeta) = \sum_{\beta \text{ satisfying } (15)} \prod_{g=1}^{r} Z_{g,\beta} \pmod{2^{p_t}}. \tag{16}$$

Here $Z_{g,\beta}$ is the contraction of the preprint's child network for group g, with the new packet coefficients. Its wires and vertex groups have not changed.

6.1 Exact arithmetic costs

Every quotient used in a feature can be evaluated by extracting the known powers of two from each numerator and denominator, then inverting only units. The net power remaining after division by 2^{E_b} is nonnegative by Lemma 4.2. Thus one need not lift a previously computed low-precision residue and divide it by a nonunit. The exact kernel prototype implements this factorwise normalization.

Alternatively, exact rational coordinates have polynomial bit length and can be reduced after cancellation. There are polynomially many factors, each initially of polynomial bit length. The scalar denominators do not depend on V, so nilpotent tensor coordinates introduce no new inversion problem. Full feature tables have polynomial size in s; they can also be streamed.

An entry of a packet tensor is a coefficient of a product of at most L affine factors, multiplied by prescribed powers of source forms. The original $\exp(O(L \log s))$ enumeration bound still applies, and the earlier streamed finite-difference improvement remains available [6]. Neither is the leading cost under the schedule below. Descriptions retain the preprint's creation-precision tags and canonical entry representatives. No recursive composition of entry evaluators is introduced.

7 Mode entropy and the polynomial-space bound

We may conservatively lower the weights to

$$\widehat{w}_q = 1 + 4q - 2s_2(q).$$

The actual base $a + 1 - i$ is at least one. We may also extend the finite range $q < D/2$ to all $q \geq 0$ for an upper bound.

Lemma 7.1 (Weighted partition estimate)**.** *As $\theta \downarrow 0$,*

$$\sum_{q \geq 0} -\ln(1 - e^{-\theta \widehat{w}_q}) = \frac{\pi^2}{24\theta} + O(\log^2(2/\theta)).$$

Consequently, if at most c modes have each weight $\widehat{w}_q$, the number $N(p, c)$ of nonnegative exponent tuples of total weight below p satisfies

$$\ln N(p, c) \leq \theta p + \frac{c\pi^2}{24\theta} + O(c \log^2(2/\theta)).$$

Proof. The sequence $\widehat{w}_q$ is increasing and $\widehat{w}_q = 4q + O(\log(q + 2))$. Its counting function is therefore $x/4 + O(\log(x + 2))$. Integrating $-\ln(1 - e^{-\theta x})$ against that counting function gives the main term

$$\frac{1}{4} \int_0^\infty -\ln(1 - e^{-\theta x})\, dx = \frac{1}{4\theta} \sum_{j \geq 1} \frac{1}{j^2} = \frac{\pi^2}{24\theta}.$$

Integration by parts bounds the discrepancy, away from the bounded initial interval, by a constant times $\int_1^\infty \theta \log(x + 2)/(e^{\theta x} - 1)\, dx$. Splitting at $1/\theta$ gives $O(\log^2(2/\theta))$; the initial interval is smaller. The generating product $\prod_q (1 - z^{\widehat{w}_q})^{-c}$ at $z = e^{-\theta}$ proves the second assertion. If c is not an integer, use the actual integer multiplicities, bounded by c, in the same inequality. □

For $c = 6M/D$, optimize the two leading terms with $\theta = (\pi/2)\sqrt{M/(Dp_t)}$. This gives

$$\ln N_\beta \leq (\pi + o(1))\sqrt{Mp_t/D}. \tag{17}$$

The error is uniform in the calls below: $p_t = (6 + o(1))MD/u$, while D grows faster than any fixed power of λ. In particular, $(M/D)\log^2(D/\sqrt{u}) = o(M/\sqrt{u})$. The analogous leading constants for slopes one and three are respectively 2π and $2\pi/\sqrt{3}$, proving the matched gain.

7.1 The schedule and every required inequality

Set $\lambda = \lceil \log_2 s \rceil$ and use

$$r = 256, \quad \rho = \frac{6}{r} = \frac{3}{128}, \quad H = \lceil s/\lambda \rceil, \quad u_d = \left\lfloor \frac{\lambda}{700 \cdot 12^d} \right\rfloor, \quad a_d = 2\sum_{j=0}^{d} u_j. \tag{18}$$

Calls with at most H edges are leaves. At depth d, set $u = u_d, a = a_d, L = 2^a, D_{\mathrm{arr}} = r^{2a}$. The geometric sum gives $a_d \le 6\lambda/1925$, so

$$L = O(s^{6/1925}), \qquad D_{\mathrm{arr}} = O(s^{96/1925}),$$
$$D_{\mathrm{arr}}^2 \left\lceil s^{9/10} \right\rceil = O(s^{3849/3850}), \qquad LD_{\mathrm{arr}} = O(s^{102/1925}).$$

The last two bounds are $o(H)$; also $2L \le \left\lceil s^{9/10} \right\rceil$ eventually. Thus the source's three packing inequalities hold at every active call. The child bound is ρM, and new packet degrees are at most $2L$, closing the graph invariants.

Since $s/\lambda < M \le s\rho^d$ at an active call,

$$u_d = \Omega(\lambda^\eta), \qquad \eta = 1 - \frac{\ln 12}{\ln(128/3)} = 0.3379616482\ldots > 0.$$

Thus all floors are uniformly negligible, $u_d \to \infty$, and $u_d/M \to 0$ throughout the active recursion.

Reuse the earlier exact taper count and valid precision allowance [7, 6]:

$$Q(a,u) = L(2u - 2 + 2^{1-u}),$$
$$h = \left\lceil \frac{M(2L-1)+1}{Q(a,u)} \right\rceil, \qquad q_t = h\max(0, u - v_2(t)), \tag{19}$$
$$p_t = p + 6hD_t + hu(a+1), \qquad D_t = L/2^{v_2(t)}. \tag{20}$$

There are enough Hermite conditions because their exact total is $hQ(a,u)$. The known Hermite multiplier loses at most $6hD_t$ powers of two; evaluating the quotient jet loses fewer than $hu(a+1)$ more. These are exact safe integer allowances, not floating-point estimates.

The inherited precision obeys $p \le s\lambda^{O(1)}2^{a_{d-1}}$. A retained node has $i < u_d$, so $D_t \ge 2^{a_{d-1}+u_d+1}$. The inherited term and the final term of (20) are therefore $o(MD_t/u_d)$ uniformly. Also $h = (1 + o(1))M/u_d$. It follows that

$$p_t = (6 + o(1))MD_t/u_d. \tag{21}$$

All precisions remain polynomial in s. The preprint's residue field of size $2^{4\lambda}$ still has more interpolation points than needed, since the deformation degree is $O(s^{1+6/1925})$.

7.2 Completing Theorem 1.1

Equations (17) and (21) give at depth d

$$\log_2 N_\beta \le (\kappa + o(1))\frac{s}{\sqrt{\lambda}}\gamma^d, \quad \kappa = \frac{\pi\sqrt{4200}}{\ln 2}, \quad \gamma = \rho\sqrt{12} = \frac{3\sqrt{12}}{128} < 1.$$

Polynomially many interpolation choices and groups per call contribute only $O(\lambda \log \lambda)$ to the overall logarithm. Summing the geometric exponents gives $\kappa/(1-\gamma) = C_{\mathrm{ps}}$. Leaf work costs $2^{O(s/\lambda)}$, and entry evaluation costs $2^{O(s^{9/10})} + \exp(O(L \log s))$ per appropriate call, both lower order than $2^{s/\sqrt{\lambda}}$. The fixed polynomial in the description length remains outside the exponential bound.

Depth-first traversal stores one tuple, one running group product and sum, the polynomially sized interpolation data, and the creation descriptions at each of $O(\log \lambda)$ levels. This proves polynomial space and Theorem 1.1.

8 A fully costed mode-graph evaluation

This section proves Theorem 1.2. It uses the same child networks as before. It does not apply the contraction theorem to the high-degree group tensors that obstruct the earlier conditional proposals.

8.1 The number of modes for each directed group pair

The packing construction pads every ordered full-label class to the same multiple H_{pad} of L. At any fixed coordinate, for distinct group symbols g, h, the number of packets with unordered pair $\{g, h\}$ is

$$\frac{2H_{\mathrm{pad}} D_{\mathrm{arr}}}{r^2 L}.$$

Indeed the two ordered symbol pairs account for exactly $2D_{\mathrm{arr}}/r^2$ full-label classes, and these packets have L slots. One orientation has source g and target h, the other the reverse. There are R modes of each index q per orientation. Since $H_{\mathrm{pad}} D_{\mathrm{arr}} \le 3M$, each directed pair has at most

$$c_{gh} \le \frac{6M}{r^2 D} \tag{22}$$

modes of each q. Real-port omissions can only reduce the required set; retaining dummy zero modes is harmless for an upper bound.

Partition each tuple β by its *exact* total weight b_{gh} on each directed pair. Enumerate nonnegative arrays with $\sum_{g \neq h} b_{gh} < p_t$. Since $r = 256$ is fixed and p_t is polynomial in s, this is a polynomial number of arrays, though with a very large fixed exponent. Every mode tuple belongs to exactly one array. For each pair, enumerate or index its finite list of tuples of that exact weight. Empty lists make the array contribute zero. This indexing does not require a counting oracle. With mode weights $w_1, \dots, w_v$, the integer table

$$D(j, b) = D(j-1, b) + D(j, b - w_j), \qquad D(0,0) = 1, \quad D(0, b) = 0 \ (b > 0),$$

with negative-budget entries zero, counts unbounded nonnegative exponent vectors of exact weight b on the first j modes. There are polynomially many modes and p_t is polynomial in s. These counts also have polynomial bit length, since they are bounded by $(p_t + 1)^v$. Thus the table, exact rank/unrank operations, and emptiness checks take polynomial resources. Backtracking only into branches with positive completion count enumerates an exact-weight list with polynomial delay, including when most candidate prefixes would be infeasible.

For one array, let h_{gh} be the base-two logarithm of the size of its pair list; ceiling it to use binary ranks adds only a constant per pair. The group factor Z_g depends only on pair indices incident with g. Its value is obtained by recursively contracting the same ρM-edge child as in (16). No inexpensive group oracle is assumed.

Lemma 8.1 (Entropy for a selected set of directed pairs)**.** *Write*

$$K(M,u)=\frac{\pi\sqrt{6}}{\ln 2}\frac{M}{\sqrt{u}}.$$

For any fixed set I of directed group pairs,

$$\sum_{e\in I}h_e\le\left(\sqrt{|I|/r^2}+o(1)\right)K(M,u). \tag{23}$$

More generally, if disjoint pair classes have cardinalities v_j and total index lengths H_j, then for fixed nonnegative w_j,

$$\sum_j w_jH_j\le\left(\sqrt{\frac{\sum_j w_j^2v_j}{r^2}}+o(1)\right)K(M,u). \tag{24}$$

Proof. Use Lemma 7.1 with the common per-pair color upper bound $c=6M/(r^2D)$. For a pair with allocated weight b_e, the generating product bounds its exact-weight count by $\exp(\theta_eb_e+cF(\theta_e))$, where F is the sum in Lemma 7.1. Assign weight w_e to that pair, and omit pairs with $w_e=0$. In its generating bound use parameter $\theta_e=\theta/w_e$. Multiplying that bound by w_e and adding gives

$$\ln 2\sum_e w_eh_e\le\theta\sum_e b_e+\frac{c\pi^2}{24\theta}\sum_e w_e^2+O(c\log^2(2/\theta)).$$

The number of pairs and their positive weights are fixed constants. This bound holds also for small or zero allocated budgets; it does not apply a large-budget saddle estimate to each pair separately. Since $\sum_e b_e<p_t$, optimize the two displayed leading terms and use $p_t=(6+o(1))MD/u$. The result is

$$\ln 2\sum_e w_eh_e\le(\pi+o(1))\sqrt{Mp_t/D}\sqrt{\sum_e w_e^2/r^2}.$$

The error is uniform for the same reason as in Section 7. This proves the two stated forms after division by $\ln 2$; fixed index rounding errors are lower order. □

8.2 Cache each group before making three block tables

Partition the 256 groups into blocks of sizes $85,85,86$. For each individual group, evaluate and store its function Z_g on all incident pair-index tuples. This pays for the recursive children explicitly. There are $2(r-1)=510$ incident directed pairs, so the number of child calls per group is at most

$$2^{(a_0+o(1))K(M,u)},\qquad a_0=\sqrt{510/65536}.$$

For a block B, form a table on all directed pairs joining it to its complement. Each entry sums over its internal pair indices the product of the cached Z_g for $g\in B$. Straight enumeration of all incident indices builds this table. For a block of size k the number of incident directed pairs and the number of boundary directed pairs are, respectively,

$$2rk-k^2-k,\qquad 2k(r-k).$$

Their maxima for $k=85,86$ are 36550 and 29240. Thus table construction has exponent coefficient

$$a_B=\sqrt{36550/65536}=0.7467989958\ldots$$

on the scale $K(M,u)$, and storing the tables has coefficient

$$j_B = \sqrt{29240/65536} = 0.6679573281\ldots.$$

The individual group caches are smaller. Work inside a block uses cached values; it does not multiply the block-table enumeration count by a new recursive child cost.

This access cost can be realized by sequential records rather than an assumed constant-time lookup into an exponential table. Enumerate a block's boundary axes first and its internal axes last, and order each group's cache by the restriction of that axis order. Projecting the full lexicographic loop onto a group's axes either repeats a record, advances, or rewinds a suffix when counters reset. A rewind crosses no more cache records than the full suffix assignments just traversed. Charging rewinds to those preceding loops bounds total tape movement by the enumeration count times the fixed number of axes and the polynomial record length. This preserves the stated bit-cost bound. The small prototype uses ordinary Python tables to check the algebra; it is not an implementation of this tape-access schedule.

Combining the two directed collections between each unordered pair of blocks into one index gives three tables on a triangle. Let their shared index dimensions be d_1, d_2, d_3 with logarithms H_1, H_2, H_3. The contraction is of the form

$$\sum_{x=1}^{d_1}\sum_{y=1}^{d_2}\sum_{z=1}^{d_3} A_{xz}B_{xy}C_{yz}. \tag{25}$$

Enumerating all crossing indices already gives coefficient $\sqrt{43690/65536}$ and time constant 239.828126, with the same space bound. The next step improves it further.

8.3 Division-free rectangular multiplication

Strassen's seven-product identity uses only addition, subtraction, and multiplication [12]. It therefore works over the current commutative ring, including modulo powers of two and with nilpotents. For rectangular dimensions d_1, d_2, d_3, split into square blocks of side comparable to $\min d_j$, pad by at most a constant factor, and apply the identity recursively to each square product. This gives arithmetic time

$$O\left(d_1d_2d_3(\min d_j)^{\omega-3}\operatorname{poly}(\log(2+\max d_j))\right), \qquad \omega = \log_2 7.$$

The remaining inner product in (25) is no larger in order. The exponent is

$$H_1 + H_2 + H_3 - (3-\omega)\min(H_1, H_2, H_3). \tag{26}$$

The three unordered block-pair classes have $14450, 14620, 14620$ directed pairs. Expression (26) is the maximum of three weighted sums, with weights $1, 1, \omega-2$. Applying (24), the worst bound discounts the smallest class, and is

$$(b_S + o(1))K(M,u), \qquad b_S = \sqrt{\frac{43690 - [1-(\omega-2)^2]14450}{65536}} = 0.7680409453\ldots.$$

Notice $a_B < b_S$. Rectangular products can be processed one square block at a time. Recursive Strassen storage for one block is quadratic in its side; the three input tables and an output table fit within the already bounded boundary-table sizes. Thus the space coefficient remains j_B. Reblocking

or transposing a stored matrix can be done by sorting its records on their new indices. The largest such record set is a boundary table; its logarithmic sorting overhead and record lengths are polynomial in s. Sequential square-block arithmetic and this reordering therefore also realize the claimed bit-operation bound without changing the space exponent.

8.4 The entire recursive time and space cost

Suppress polynomial factors per call. A correct upper recurrence is

$$T_d \le 2^{(a_0+o(1))K_d}\, rT_{d+1} + 2^{(a_B+o(1))K_d} + 2^{(b_S+o(1))K_d}, \tag{27}$$

where $K_d \le (\kappa + o(1))s\gamma^d/\sqrt{\lambda}$. The first term pays for all cached group functions; the other two build and contract the tables. Exact weight arrays and interpolation choices multiply the recurrence by polynomial factors.

At depth d, the coefficient from the preceding cached-child enumerations and the current matrix work is at most

$$a_0 \frac{1-\gamma^d}{1-\gamma} + b_S\gamma^d.$$

It is bounded by $\max\{a_0/(1-\gamma), b_S\} = b_S$, because $a_0/(1-\gamma) = 0.0960106375\ldots < b_S$. The a_B term and leaves are smaller. Hence the full time coefficient is $\kappa b_S = C_{\text{fast}}$, proving the time assertion.

The largest stored tables at depth d have exponent at most $j_B K_d$. Ancestor caches and tables have polynomially many levels and geometrically decreasing exponent bounds, so their total has leading coefficient $\kappa j_B = C_{\text{mem}}$. Polynomial-bit ring entries and their arithmetic costs do not change either coefficient. This proves Theorem 1.2.

Why this does not prove the old conditional targets. The pair-index representation was already suggested in the earlier packets. Here the group tensors are evaluated by explicit tables and ordinary matrix multiplication. Their work remains exponential in $M/\sqrt{u}$ and their storage is subexponential in the original network size. We have not supplied a polynomial-space, subexponential-in-the-mode-count oracle or a low-degree factorization of the high-degree group tensors. The conditional $s/\log s$ targets do not follow.

9 Consequences for SAT and the limits of the result

At fixed clause density, the source builds a bounded-degree Boolean network with $O_{k,D}(n)$ wires. At unrestricted density, its sparse parity-query reduction uses $O_k(n \log\log n)$ wires. Substitute either new engine without changing those encodings. Theorem 1.1 gives the same orders (1)–(2), with an improved explicit contraction certificate. Theorem 1.2 gives a further time coefficient reduction while using space of the same subexponential order as its time.

The density-sensitive reduction of the reviewed packets can likewise be retained. With the polynomial-space engine, its established envelope remains

$$(n+m)^{O_k(1)} 2^{O_k\left(\frac{n}{\sqrt{\log n}} \min\{\log(2+m/n), \log\log n\}\right)}.$$

The scalar parity lift, exact recovery budgets, and streamed coefficient extraction are compatible with the new modes. Their results are inherited improvements, not new contributions of this note.

9.1 What was not obtained

This investigation does not prove a smaller asymptotic order for the worst-case exponent in n. Both new engines still have $\log T = \Theta_{\mathrm{bound}}(s/\sqrt{\log s})$ as an upper-bound scale; the notation here describes the scale of the certificate, not an algorithmic lower bound. Neither $2^{O(n/\log n)}$ nor a removal of the dense-case $\log\log n$ factor has been established.

The exact Smith profile explains why a further integral basis change in this same full block cannot turn its linear valuation growth into a quadratic profile. It does not rule out cancellation across modes, a more economical reduction, a different kernel, or a different algorithm. The fast option demonstrates that some useful sharing is possible without the earlier structural oracle, but only yields a constant improvement on the present entropy scale.

9.2 A numerical comparison of certificates

Construction or comparison	Time coefficient	Space
Published schedule, audited geometric count	18535.7538	polynomial
Earlier improved explicit schedule	648.4411	polynomial
Earlier schedule plus earlier factorial modes	374.3776	polynomial
Earlier combined schedule-family infimum	368.6540	polynomial
New full-block modes on the same explicit schedule	324.2205	polynomial
New full-block modes, taper scale 1/700	319.6858	polynomial
New three-block tables, direct triangle sum	239.8281	exp., 196.1994
New cached groups and Strassen triangle sum	225.5971	exp., 196.1994

Every numerical entry refers to the normalized $s/\sqrt{\log_2 s}$ scale. An exponential-space entry is its coefficient on that same scale. The infimum row is a limiting certificate, not an attained algorithm at the packing boundary. The first row is a later explicit reanalysis of the source parameters, not a numerical theorem stated by the source. These numbers are not practical benchmarks.

The rounded bounds 320, 226, and 197 do not depend on floating-point rounding. To certify them, use

$$\pi < \frac{22}{7}, \qquad \ln 2 > \frac{6931}{10000}, \qquad \gamma < \frac{13}{160}, \qquad \omega < \frac{281}{100}.$$

The logarithm lower bound follows from four terms of $\ln 2 = 2\sum_{j\ge 0} 3^{-(2j+1)}/(2j+1)$; the last inequality follows from $7^{100} < 2^{281}$. Substitute these rational bounds into the squared coefficient formulas. Exact rational arithmetic gives values strictly below 320^2, 226^2, and 197^2, respectively. The comparison program records these checks and the rational margins.

10 Executable artifacts, checks, and integration

The archive contains this report as `report.tex` and `report.pdf`, original standard-library Python prototypes, exact validation records, a proposal inventory, and source hashes. The proofs above establish the all-size conclusions; finite tests check the implemented identities and look for counterexamples to their stated finite instances.

10.1 Kernel validation

The kernel is implemented both over rational scalars and over $\mathbb{Q}[U]/(U^2+U+1)$. The latter supplies an actual unramified extension whose nontrivial residue classes are not ordinary integers. The checker evaluates the original cancelled polynomial independently of the separated form, checks both orientations, and tests exact and modular contractions. Each of the normal and optimized Python runs passed the following checks:

Check	Count per run
Kernel configurations	156
Exact original-polynomial reconstructions	10,248
Modular truncated reconstructions	25,620
Integral source features	5,124
Integral normalized row features	5,124
Certified target-weight inequalities	5,124
Unit diagonal checks	1,584
Triangular support checks	2,166
Exact minors evaluated	414
Verified determinantal divisors	24
Exact two-child packet contractions	4
Truncated two-child packet contractions	24

There are 55,042 successful checked conditions, in addition to 414 minor evaluations, in each run. Both runs have identical substantive results. These tests include missing slots, both crossing orientations, non-prime extension values, and the zero-prefactor case.

A separately written verifier passed 656,768 explicit conditions in each normal and optimized run. It checked 180 rational factorizations and 180 unramified-extension factorizations, with 19,116 matrix entries in each family, plus 436 small minors. Its substantive results also agree between the two runs. This additional implementation checks the normalized block factorization independently of the production prototype's formulas.

10.2 Mode-graph validation and reproduction

The second prototype contracts finite directed mode graphs by literal tuple summation and by cached three-block tables. It uses exact per-pair weight allocations so every globally feasible tuple is counted once. The checker also compares division-free rectangular Strassen multiplication with direct integer and modular products. Its complete test counts and numerical certificate calculations are recorded in the supplied JSON results. Each normal and optimized run passed 312 graph-sum comparisons, 208 cache-call checks, 114 empty-domain checks, 24 rectangular products, and 24 triangle contractions. The tests include directed reverse pairs, missing pairs, unequal blocks, repeated state weights, negative entries, and residues modulo two and sixteen. The exact-weight indexing component separately passed 119 budget cases, 366 rank/unrank round trips, and three invalid-input checks. Twelve abstract colored-partition models are counted exactly in the supplied comparison CSV; those counts are not SAT runtimes.

From the archive directory, use:

```
python3 code/verify_joint_cauchy.py
python3 code/verify_mode_graph.py
python3 code/verify_joint_cauchy_independent.py
python3 code/compare_certificates.py
python3 build_report.py
```

The first four commands use only the Python standard library. The final command requires a standard LaTeX installation with the packages listed in the source preamble. Tests use explicit exceptions rather than assertions that disappear under `python -O`. No upstream executable or Lean development was run during this investigation.

10.3 Integration points

In the preprint's `quantitative-contraction.tex`, replace the geometric crossing expansion by Theorem 4.3; replace its weight and partition-count estimates by (11) and (17); retain its local kernel, Hermite identity, child network topology, and creation-precision convention. Use (18)–(20) for the explicit coefficient in Theorem 1.1. The polynomial-space implementation continues to stream feasible tuples.

For Theorem 1.2, replace the streaming sum in (16) by the exact weight-array loop, individual group caches, three block tables, and rectangular multiplication of Section 8. This is an optional branch because it changes the space guarantee.

The delivered programs implement and check these local constructions. They do not include the preprint's complete packing, global interpolation, formula sparsification, or parity-query search engine. The main imported proofs and the claimed global ETH result have not been independently formalized here. The report and artifacts are a reviewable mathematical refinement, with those dependencies stated explicitly.

A A short arithmetic specification for the seven-product step

For square block matrices $A = (A_{ij})$ and $B = (B_{ij})$, define

$$\begin{aligned} M_1 &= (A_{11} + A_{22})(B_{11} + B_{22}), & M_2 &= (A_{21} + A_{22})B_{11}, \\ M_3 &= A_{11}(B_{12} - B_{22}), & M_4 &= A_{22}(B_{21} - B_{11}), \\ M_5 &= (A_{11} + A_{12})B_{22}, & M_6 &= (A_{21} - A_{11})(B_{11} + B_{12}), \\ M_7 &= (A_{12} - A_{22})(B_{21} + B_{22}). \end{aligned}$$

Then

$$AB = \begin{pmatrix} M_1 + M_4 - M_5 + M_7 & M_3 + M_5 \\ M_2 + M_4 & M_1 - M_2 + M_3 + M_6 \end{pmatrix}.$$

These are polynomial identities with integer coefficients. They remain valid modulo 2^p without an inverse of two and without a field assumption. Recursing gives exponent $\log_2 7$. A depth-first schedule retains $O(d^2)$ ring elements for a square block of side d; summing its geometric storage across recursion levels preserves that order. This supplies the specific matrix multiplication guarantee used in Theorem 1.2.

B Proof dependencies

Result	Dependency status
Full crossing separation and integral factors	Proved in this report by finite telescoping and ordinary integer factorial divisibility.
Full padded Smith profile	Proved by the displayed unit triangular factorization; agrees with the earlier $i = 0$ calculation.
Weighted entropy and pair allocation bounds	Proved here from the generating product and Cauchy–Schwarz.
Taper count, recovery identity, creation-precision rules	Imported from the preprint and prior packets; parameter compatibility is checked here.
Packing and child contraction semantics	Imported from the preprint; group support and new multiplicities are checked here.
Three-block caching and rectangular combination	Proved here using exact finite summation and the displayed seven-product identity.
General and density-sensitive SAT reductions	Imported; no new reduction claim or smaller logarithmic order is made.

References

[1] OpenAI. *The Exponential Time Hypothesis is False*. October 25, 2026 revision. Pinned preprint directory.

[2] OpenAI. `quantitative-contraction.tex`, same snapshot. Exact contraction source.

[3] OpenAI. `quantitative-reduction.tex`, same snapshot. Exact reduction source.

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[5] *A factorial Cauchy expansion for the subexponential 3-SAT algorithm*. Research packet, October 26, 2026. Report source.

[6] *Quantitative Improvements to the Exact-Counting Contraction Algorithm*. Research packet, October 26, 2026. Report source.

[7] *3SAT precision refinement*. Research packet, October 26, 2026. Report source.

[8] *A density-sensitive refinement of the October 2026 exact k-SAT counting algorithm*. Research packet, October 26, 2026. Report source, especially the local Smith calculation.

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[12] Volker Strassen. Gaussian elimination is not optimal. *Numerische Mathematik* 13 (1969), 354–356. doi:10.1007/BF02165411.